

DETERMINATION OF COEFFICIENT OF DISCHARGE AND CALIBRATION OF A RECTANGULAR NOTCH

AIM

1. Determine the coefficient of discharge, C_d of the given rectangular notch.
2. Plot the graph Q_a Vs H .
3. Plot the graph Q_a Vs $H^{3/2}$, find the value of ' C_d ' from the graph.
4. Calibrate the notch and determine the actual discharge for a head of.....cm.

APPARATUS REQUIRED

1. Tank with a rectangular notch
2. Hook gauge
3. Collecting tank with piezometer.
4. Stop watch

PRINCIPLE

Coefficient of discharge,

$$C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}} = \frac{Q_a}{Q_{th}}$$

Actual discharge,

$$Q_a = \frac{A \times R}{t} \frac{\text{cm}^3}{\text{s}}$$

Where

A = Area of collecting tank in cm^2

R = Height of water collected in cm

t = Time for collection for ' R ' centimeters of water rise in seconds

Theoretical discharge,

$$Q_{th} = \frac{2}{3} L \sqrt{2g} H^{3/2} \frac{\text{cm}^3}{\text{s}}$$

Where

L = Width of the notch in cm.

H = Head above the sill of the notch = $(h_2 - h_1)$ in cm

h_1 = Sill level reading in cm

h_2 = Final hook gauge reading in cm

g = Acceleration due to gravity in cm/s^2

PROCEDURE

1. The internal plan dimensions of the collecting tank and the width of the notch are measured.
2. Open the inlet (supply) valve and close the bypass valve.
3. Start the motor and the water is allowed to rise only up to the still level of the notch, supply valve is closed tightly and switch off the machine.
4. The tip of the pointer of the hook gauge is adjusted such that the tip just touches the free water surface.
5. The reading at sill of the notch (h_1) is noted from the hook gauge.
6. Start the pump, the supply valve is opened, the bypass valve is partly open and the water is allowed to flow through the notch. The tip of the pointer is adjusted to coincide with the water surface.
7. The reading in the hook gauge (h_2) is noted.
8. The out let valve of the collecting tank is tightly closed and the time (t) taken for a known rise (R) in the collecting tank is measured.
9. Similarly the above procedure is repeated by gradually increasing the heads of the flow by partly closing the bypass valve. (Minimum 6 sets)
10. The observations are tabulated and the coefficient of discharge is calculated.

OBSERVATIONS AND TABULATION						
SL.No	Hook gauge readings				Time for 'R' cm rise, t s	Actual discharge, Q_a cm^3/s
	Sill level reading, h_1 cm	Final hook gauge reading, h_2 cm	Head over the notch, H cm	$H^{3/2}$ cm		
Unit	cm	cm	cm	cm	s	cm^3/s
1						
2						
3						
4						
5						
6						

Coefficient of
discharge, Cd

Theoretical
discharge, Q_{th}
 cm^3/s

SAMPLE CALCULATION { SET No. }

Area of collecting tank, A	=	cm ² .
Rise of water, R	=	cm
Time for collection, t	=	s
Actual discharge, $Q_a = \frac{A \times R}{t} \frac{\text{cm}^3}{\text{s}}$	=	cm ³ /s
Breadth of notch, L	=	cm
Acceleration due to gravity, g	=	981 $\frac{\text{cm}}{\text{s}^2}$
Sill level reading, h ₁	=	cm
Final hook gauge reading, h ₂	=	cm
Head above the notch, H, (h ₂ -h ₁)	=	cm
Theoretic al discharge, $Q_{th} = \frac{2}{3} L \sqrt{2g} H^{3/2}$	=	$\frac{\text{cm}^3}{\text{s}}$
Coefficient of discharge, $C_d = \frac{\text{Actual discharge}}{\text{Theoretical discharge}}$	=	

CALIBRATION

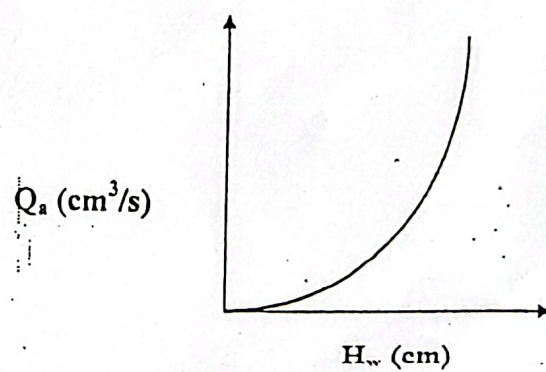
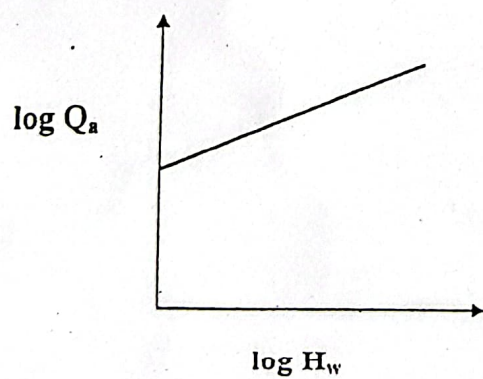
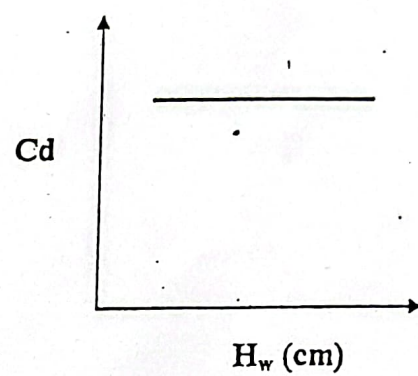
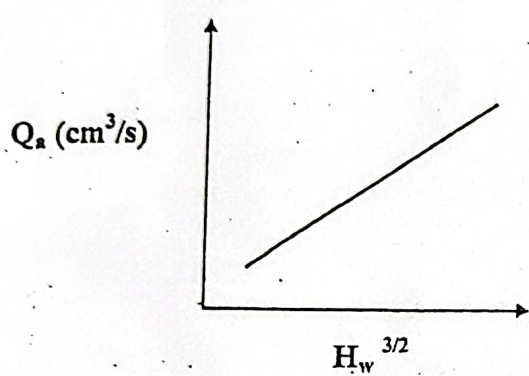
Actual discharge through the rectangular notch is assumed to be governed by the equation,

$$Q_a = KH^n$$

Where

Q_a	=	Actual discharge
H	=	Head over the notch, K and n are Constants

Model graphs for Rectangular Notch



Taking log on both sides

$$\log Q_a = \log K + n \log H$$

Plotting $\log Q_a$ Vs $\log H$ graph, the slope of the line gives the value of n and the intercept on the $\log Q_a$ axis gives $\log K$, Substituting these values of K and n in the equation for discharge $Q_a = KH^n$, the calibration equation can be obtained.

Q_a can be calculated for suitable values of H . The calibration curve can be plotted with H on X-axis and Q_a on Y-axis.

RESULT

Coefficient of discharge of the rectangular notch, $C_d =$

INFERENCE

GRAPHS

- | | | | |
|----|-----------|-------|-------------------------|
| 1. | Q_a | V_s | H |
| 2. | Q_a | V_s | $H^{3/2}$ |
| 3. | $\ln Q_a$ | V_s | $\ln H$ |
| 4. | Q_a | V_s | H – Calibration graph |